

RUTHERFORD COUNTY AP, NC, USA

WMO: 723144

Lat: 35.428N

Lon: 81.935W

Elev: 1078

StdP: 14.13

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53890

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	24.1	-2.2	5.1	34.3	2.7	6.6	34.7	18.6	47.8	16.3	46.6	1.7	350	0.354

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	92.8	72.9	90.5	73.3	88.2	72.4	77.2	85.6	76.1	84.2	75.2	82.9	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.0	136.8	79.6	73.4	129.5	78.8	72.9	126.9	78.3	41.3	85.3	40.2	83.9	39.4	83.2	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.4	10.6	11.8	94.3	5.0	3.2	8.2	96.6	5.2	98.4	2.4	100.2	-1.2	102.5 (4)		
(5)			10.4	79.0	4.2	1.2	7.3	79.9	4.9	80.6	2.5	81.3	-0.5	82.2 (5)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	41.3	44.7	51.0	60.1	67.5	74.9	77.4	76.7	71.4	61.0	51.3	44.2
	DBStd	14.67	9.77	8.53	8.68	7.48	5.92	4.20	4.07	3.82	5.38	7.48	8.00	8.47
	HDD50	908	305	188	96	11	0	0	0	0	0	9	77	222
	HDD65	3208	735	568	439	181	46	1	0	0	14	164	414	646
	CDD50	4630	35	41	127	313	543	747	848	828	641	350	116	41
	CDD65	1456	0	1	5	32	124	299	383	363	205	40	3	1
	CDH74	12375	4	7	77	419	1093	2575	3365	2882	1559	352	35	7
	CDH80	4284	0	0	9	70	268	939	1379	1063	498	56	1	1
Wind	WSAvg	3.3	3.5	3.8	4.2	4.6	3.6	2.9	2.5	2.3	2.8	3.0	3.0	2.9
Precipitation	PrecAvg	52.7	4.8	4.2	5.4	4.5	3.7	4.7	4.9	4.8	4.6	3.5	3.6	4.0
	PrecMax	69.4	8.2	7.4	8.3	9.6	8.9	8.7	9.8	8.4	14.9	11.9	8.9	6.9
	PrecMin	38.0	1.8	1.7	1.9	1.3	1.9	0.9	2.4	0.6	1.1	0.0	1.2	1.8
	PrecStd	9.0	2.0	1.7	1.9	2.4	1.8	2.5	1.8	2.4	3.2	3.1	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	73.5	81.2	85.6	89.8	93.5	97.2	94.1	91.5	84.6	78.9	72.5
		MCWB	57.7	59.9	60.6	64.6	69.5	71.8	73.3	72.8	70.4	69.0	61.3	59.5
	2%	DB	66.0	68.6	75.3	81.7	85.8	91.1	93.0	91.2	89.2	81.2	73.2	66.5
		MCWB	56.5	55.8	57.8	62.1	68.1	72.9	73.4	74.6	70.0	65.3	59.1	59.4
	5%	DB	62.6	63.9	71.9	79.2	82.4	88.3	90.5	88.9	85.6	78.7	70.4	63.1
		MCWB	54.4	53.1	56.6	61.2	67.1	72.3	74.3	73.7	70.5	64.1	58.2	54.0
	10%	DB	57.2	60.7	66.5	75.3	80.8	85.0	87.9	85.9	82.2	74.5	65.8	58.7
		MCWB	49.4	50.8	53.9	60.0	66.5	71.3	73.8	72.6	69.2	63.4	56.7	51.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.2	64.2	64.1	69.1	73.4	77.2	78.5	78.4	75.4	73.9	68.9	66.3
		MCDB	67.4	69.3	73.5	77.2	80.3	85.4	87.7	87.0	82.2	78.3	72.5	69.5
	2%	WB	59.8	60.9	61.5	66.8	71.7	75.6	77.2	77.3	74.1	70.5	64.3	61.2
		MCDB	64.4	65.2	70.1	75.6	79.2	84.6	86.0	85.5	80.8	76.1	68.1	64.7
	5%	WB	56.5	56.5	59.4	65.0	70.5	74.5	76.2	76.2	72.7	68.0	61.6	57.2
		MCDB	61.0	61.6	67.5	73.5	78.2	83.3	84.7	83.7	79.5	74.6	66.3	60.6
	10%	WB	51.6	52.3	56.8	62.9	69.0	73.4	75.2	75.2	71.6	65.4	58.8	53.1
		MCDB	55.1	57.6	64.2	70.7	76.7	81.9	83.4	82.0	78.5	72.2	64.6	56.7
Mean Daily Temperature Range	5% DB	MDBR	19.9	20.7	21.9	22.9	21.3	19.8	18.9	17.7	18.9	22.1	22.9	19.4
		MCDBR	24.7	26.8	28.4	28.2	24.9	24.3	23.4	21.9	24.1	25.5	28.1	24.9
		MCWBR	15.8	16.4	14.7	12.8	10.7	9.0	8.4	7.9	8.5	12.0	15.2	15.9
	5% WB	MCDBR	19.5	21.5	21.5	22.1	20.0	19.5	19.6	17.6	17.8	18.7	19.2	19.0
		MCWBR	14.4	16.0	13.4	11.8	9.9	8.6	8.5	7.5	8.3	11.2	13.6	15.2
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.345	0.393	0.446	0.475	0.520	0.512	0.409	0.366	0.323	0.304	
	taud	2.524	2.492	2.435	2.300	2.177	2.134	2.037	2.039	2.312	2.420	2.503	2.554	
	Ebn at Noon	286	296	293	283	268	259	246	246	269	272	276	280	
	Edh at Noon	26	30	34	41	47	49	54	53	38	31	26	23	
All-Sky Solar Radiation	RadAvg	829	1034	1363	1691	1872	1978	1842	1711	1467	1217	919	712	
	RadStd	49	113	91	129	147	140	128	81	133	137	79	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+297	+191

Nomenclature: See separate page